

Lab - 2: Investigating Network Traffic over HTTP vs HTTPS

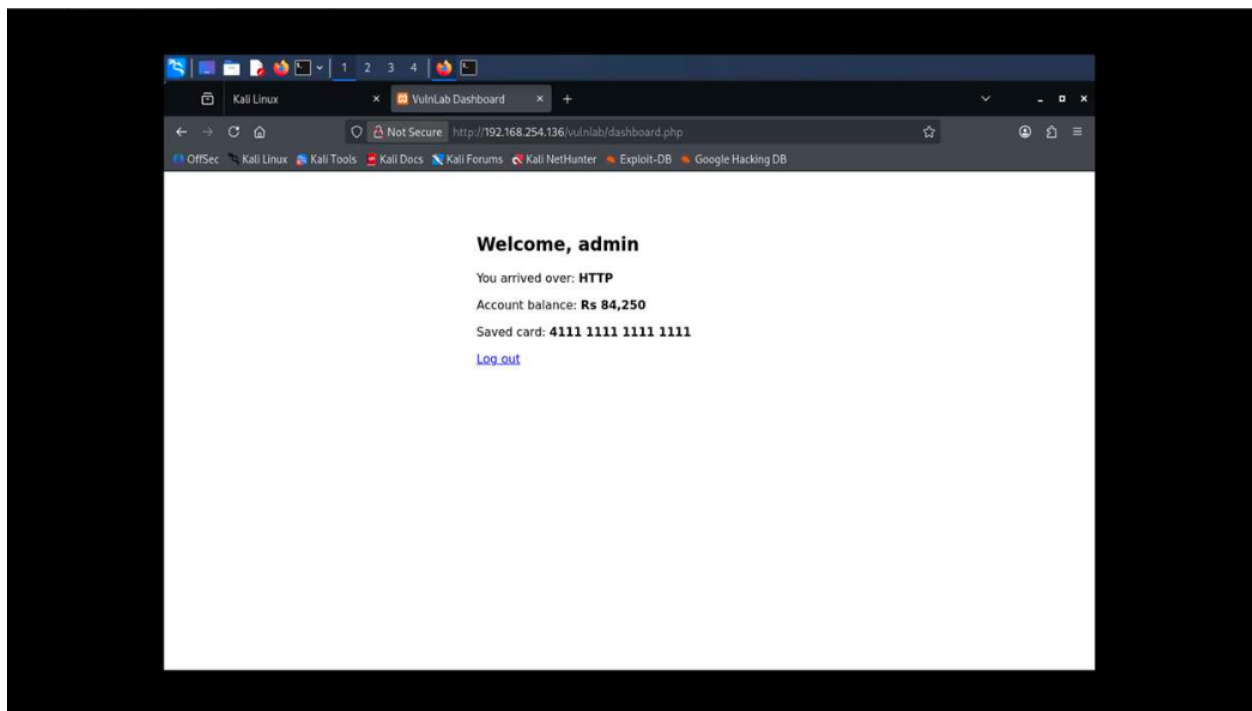
Objective:

To examine and compare how sensitive information (login credentials) is transmitted through unencrypted (HTTP) and encrypted (HTTPS) protocols using Wireshark.

Phase 1: HTTP Traffic Analysis (Unencrypted)

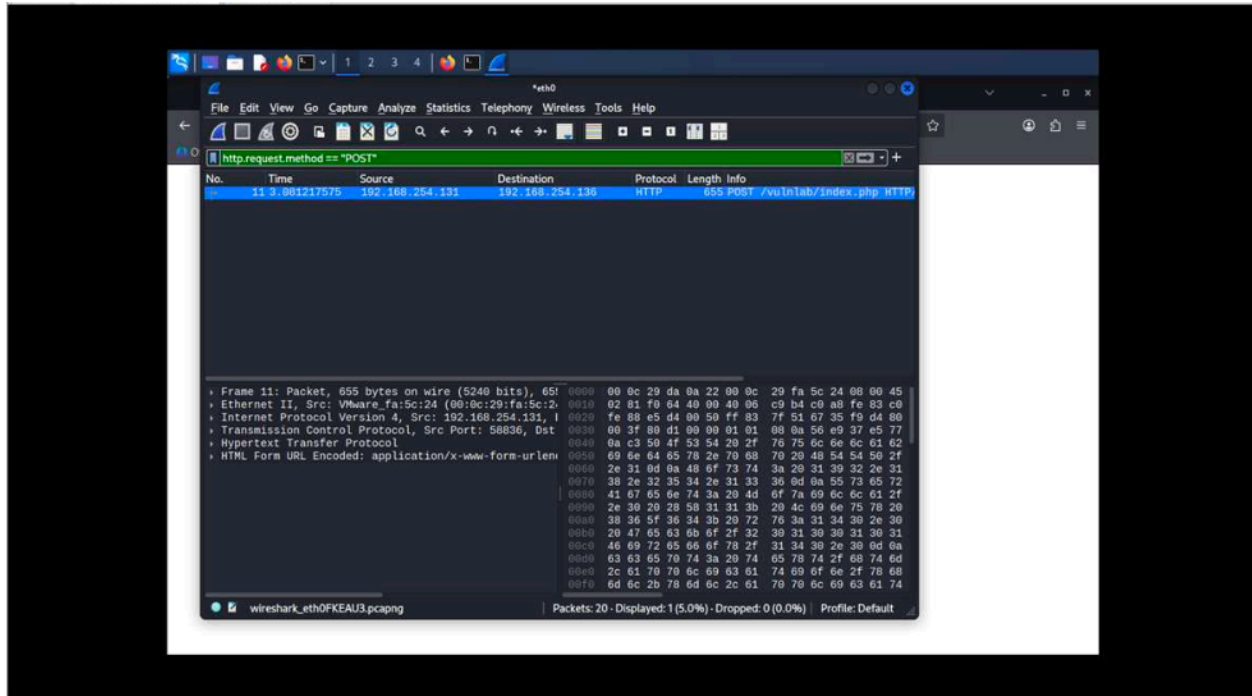
1. The Login Process

The analysis starts with a user opening the "VulnLab Bank Login" portal through a regular HTTP connection. As shown in login.png, the browser clearly indicates that the connection is "Not Secure." The user provides the username admin along with a password and successfully accesses the dashboard, confirming that the request was made over HTTP and exposing sensitive financial information.



2. Packet Capture and Analysis

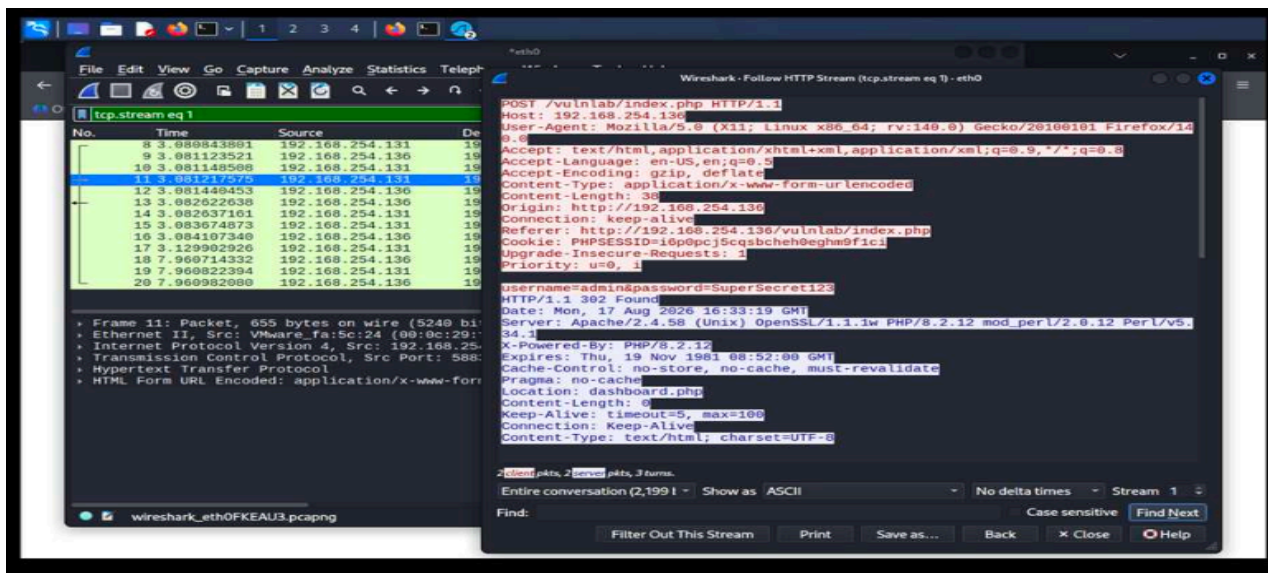
Wireshark is used to capture and inspect the network traffic, with a filter applied to identify HTTP POST requests. This helps pinpoint the exact time when the user sends their login credentials to the server.



3. The Vulnerability

By following the TCP stream of the captured HTTP POST request (follow http stream.jpg), the major security weakness of HTTP becomes apparent:

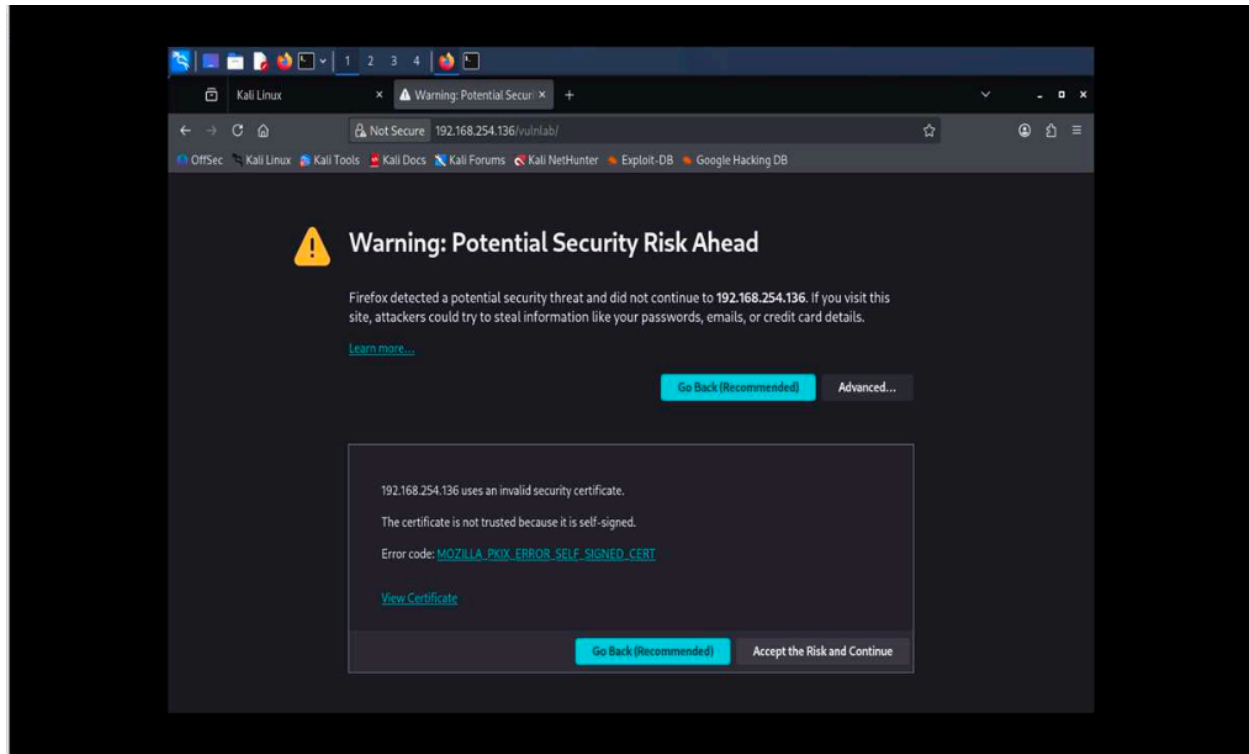
- Plaintext Transmission: The complete data payload is transmitted without encryption.
- Exposed Credentials: The text `username=admin&password=SuperSecret123` is clearly visible to anyone observing the network traffic. An attacker carrying out a Man-in-the-Middle (MitM) attack could easily obtain these login credentials.



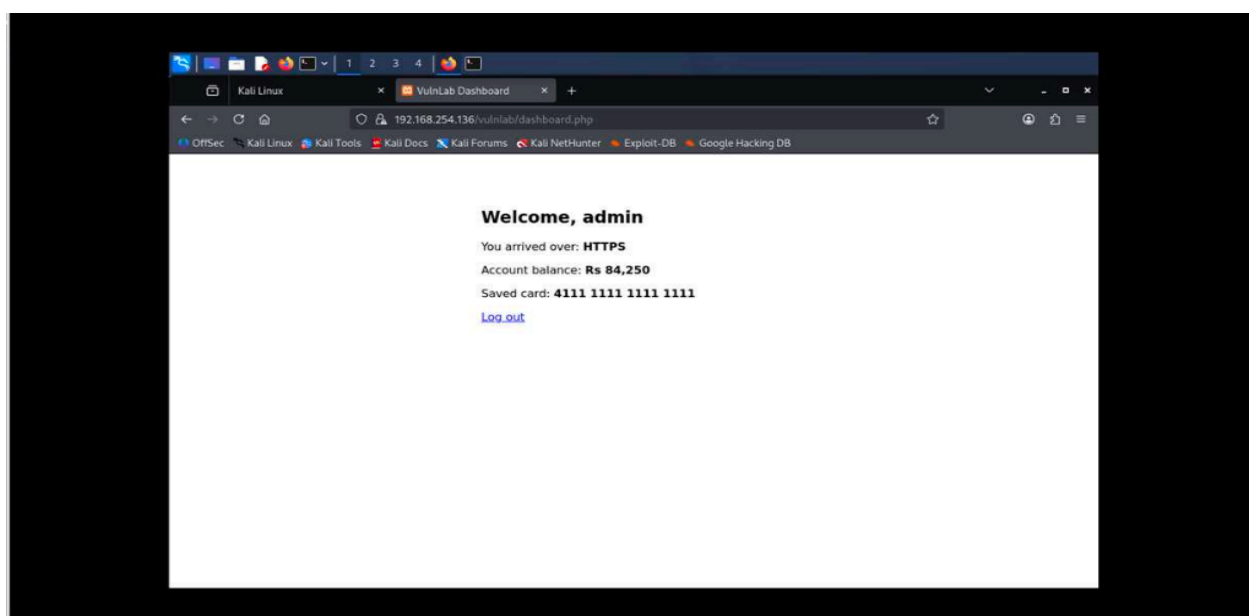
Phase 2: HTTPS Traffic Analysis (Encrypted)

1. Establishing a Secure Connection

The user next accesses the same portal using HTTPS. Since the lab setup relies on a self-signed certificate instead of a certificate issued by a trusted Certificate Authority, the browser displays a "Potential Security Risk Ahead" warning.

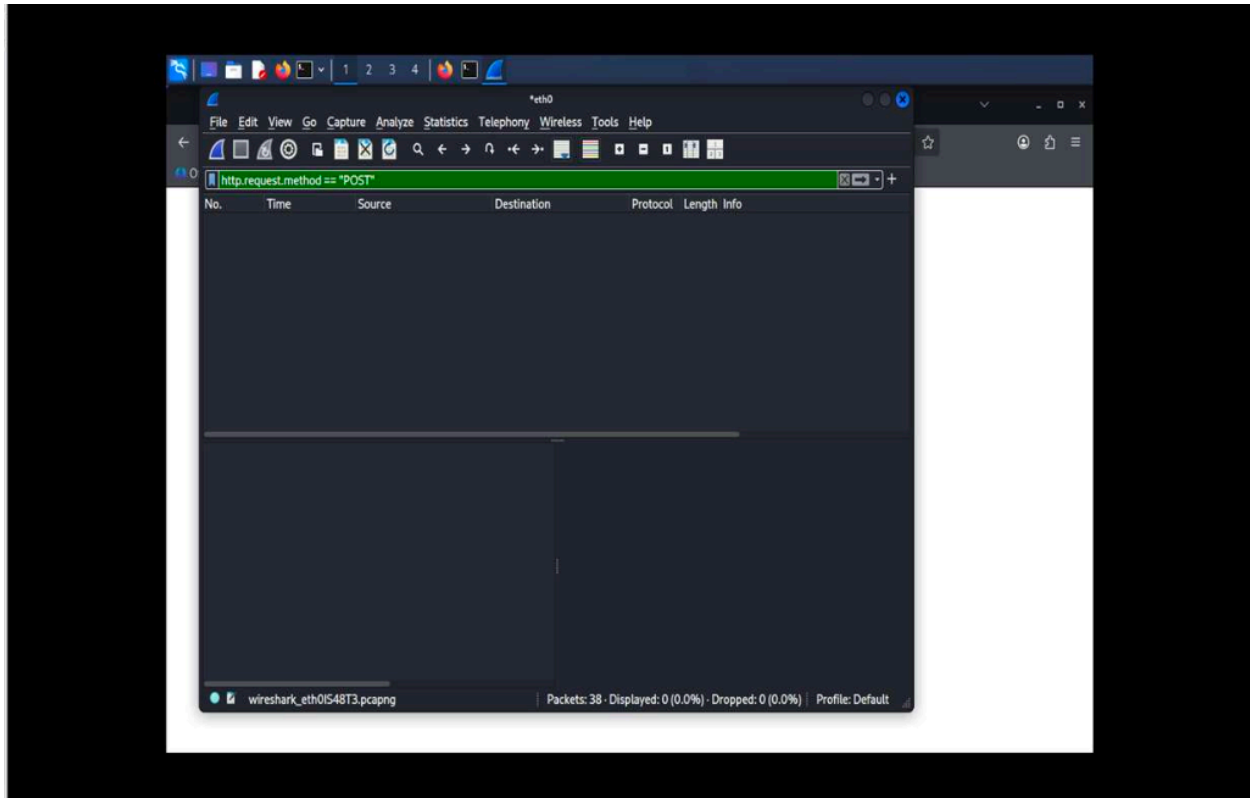


After accepting the risk, the user logs in and reaches the secure dashboard, confirming the arrival over HTTPS.



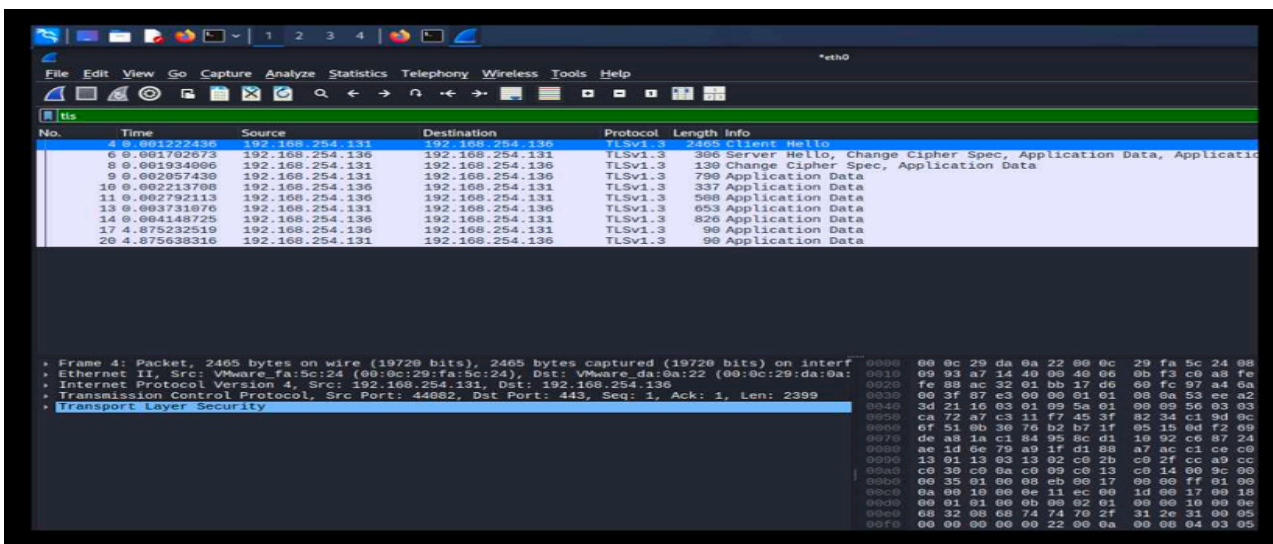
2. Packet Capture and Analysis

When the Wireshark analysis is repeated for this new session, using the same `http.request.method == "POST"` filter produces no results (second wireshark.png). This occurs because the HTTP data is now protected inside an encrypted secure tunnel.

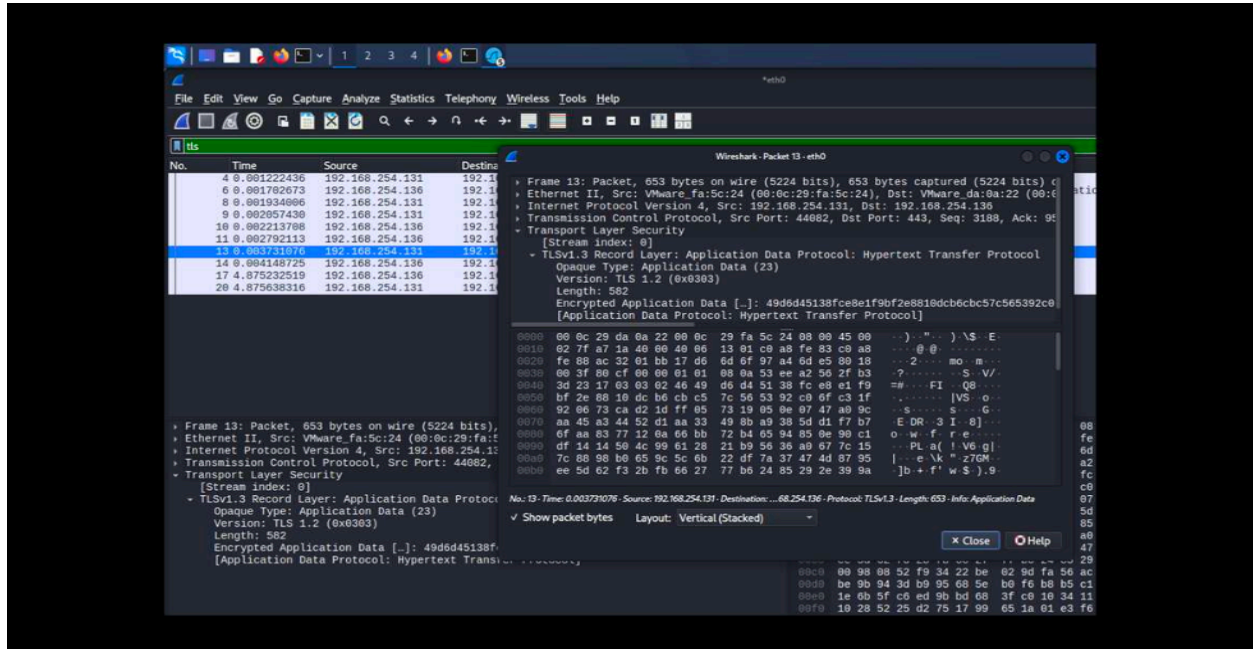


3. The TLS Handshake and Encryption

Instead of raw HTTP, Wireshark captures Transport Layer Security (TLS) traffic. This shows the client and server negotiating encryption parameters (Client Hello, Server Hello, Change Cipher Spec, Application Data, Application Data).



Encrypted Payload: When examining the Application Data packets, the contents are fully unreadable. Wireshark identifies the payload as Encrypted Application Data.



Data Protection: To verify the credentials are truly hidden, a Wireshark search for the string frame contains "SuperSecret123" returns zero packets. The encryption successfully masks sensitive data.

